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HeatPath–GNN

Today I will explain a particle-allocation problem. We have a fixed packed bed and a limited amount of highly conductive material. The question is where to place that material to improve heat transfer. We first solve and optimize a physical thermal network, then train a graph neural network to propose allocations for another packing. The results cover thirty independent realizations of one packing specification.

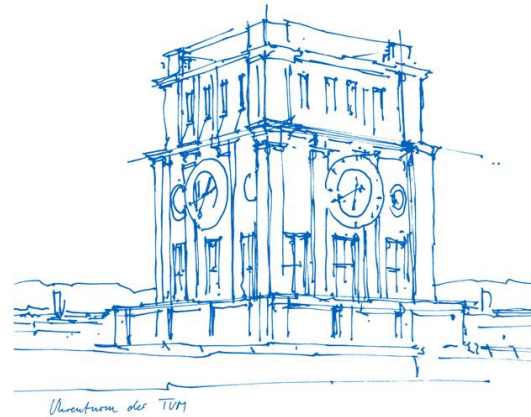
HeatPath–GNN

Learning where to place conductive particles

Thermal transport through particle contact networks

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Motivation: conductive material is a limited resource

Heat transfer through particle beds

- Heating and cooling granular products
- Thermal storage and temperature control
- Transport depends on contacts between particles

A design choice

- Highly conductive material may be costly
- The material budget is fixed
- Its position within the bed can change the conducting network

Can we obtain better heat transfer from the same amount of conductive material?

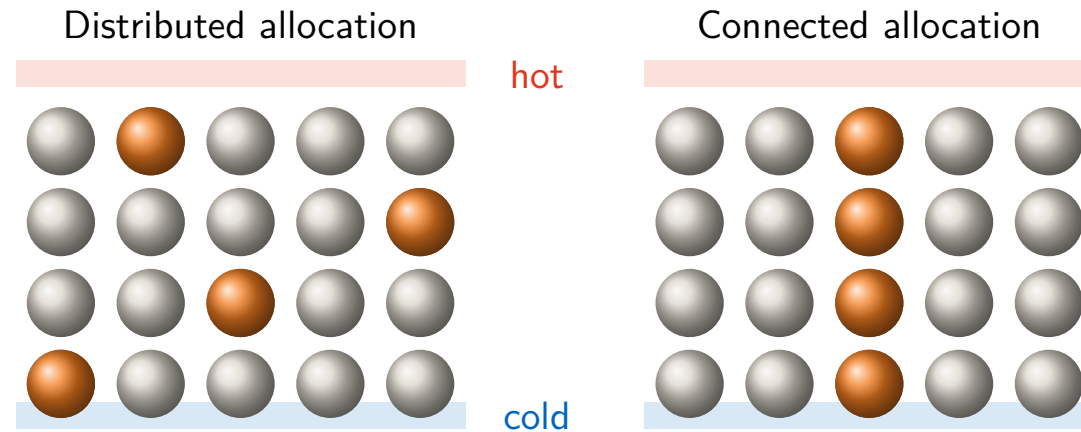
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Motivation: conductive material is a limited resource

The motivation is to improve transport without increasing the conductive-material budget. An effective conductivity describes the whole bed, but it hides the individual contacts through which heat moves. The design variable here is the assignment of material to particles. We keep the geometry fixed while comparing assignments, so any difference between methods is attributable to allocation within this model. Practical manufacturing of a prescribed assignment remains a separate question.

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The allocation problem



Orange: high conductivity. Grey: low conductivity. Schematic only.

Geometry and material amount stay fixed. Only particle labels change.

The allocation problem

This schematic illustrates the hypothesis, not a measured packing or an optimal solution. Both arrangements contain the same number of orange particles. Connecting conductive particles could reduce resistance, but the whole network, including interfaces with low-conductivity particles, determines the result. The actual problem has five hundred particles and three size classes. We enforce the budget separately within each class, which also fixes the high-conductivity solid volume.



Thirty packings with the same nominal porosity

Quantity	Value
Particles	500
Diameters [mm]	1.5 / 2.0 / 2.5
Counts	100 / 300 / 100
Nominal porosity	0.38
Independent seeds	30

- Periodic boundaries in x and y
- Two fixed thermal walls in z
- No gravity during generation
- Grow small spheres, then relax
- Accept final translational $KE \leq 10^{-12}$ J

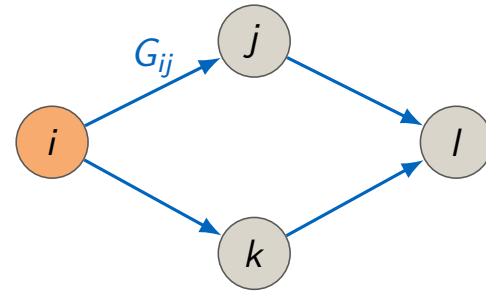
DEM: Hertz–Mindlin, $E = 50$ MPa, $\nu = 0.25$, restitution 0.30, friction 0.50.

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Thirty packings with the same nominal porosity

Each realization contains the same exact size distribution and occupies the same fixed-volume cell. Nominal porosity is calculated from the sum of sphere volumes divided by cell volume. With overlapping DEM spheres, this nominal quantity is not exactly the geometrical void fraction of their union. The modulus is deliberately softened. The final packings pass the particle-count, diameter, porosity, kinetic-energy and wall-spanning checks. We vary the random initialization, not the physical specification.

A contact network represents steady heat conduction



Nodes: particles Edges: contacts

Conservative balance at each particle

$$\sum_j G_{ij}(T_i - T_j) + \sum_w G_{iw}(T_i - T_w) = 0$$

$$k_{\text{eff}} = \frac{QH}{A(T_{\text{hot}} - T_{\text{cold}})}$$

Q : wall heat rate, H : bed height, A : wall area.

Solve particle temperatures, then calculate the bed conductivity.

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A contact network represents steady heat conduction

Each particle is represented by one temperature. For every contact, the heat leaving one particle enters its neighbour with the opposite sign. Summing the nodal balances therefore cancels internal heat exchanges. The boundary heat rate defines the effective conductivity. Hot and cold wall temperatures are 301 and 300 kelvin. This is a steady contact-conduction model. It excludes fluid-gap conduction, radiation and convection, and it does not resolve temperature variations inside each particle.

Thermal conductance follows contact size and material

Particle–particle contact

$$a_{ij} = \sqrt{R_{ij}^* \delta_{ij}}, \quad R_{ij}^* = \frac{R_i R_j}{R_i + R_j}$$

$$G_{ij} = \frac{4a_{ij}k_i k_j}{k_i + k_j}$$

δ_{ij} : geometric overlap.

a_{ij} : Hertz contact radius.

Ideal circular constriction model. Its small-contact assumption needs physical assessment.

Material assignment

$$k_i \in \{1, 10\} \text{ W m}^{-1}\text{K}^{-1}$$

Isothermal wall

$$G_{iw} = 4a_{iw}k_i$$

Two spreading resistances in series give the particle-contact formula.

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Thermal conductance follows contact size and material

The contact radius comes from the Hertz geometrical relation between overlap and effective radius. The thermal model then adds two half-space spreading resistances, each equal to one over four $k a$. For identical conductivities this gives G equals two $k a$. An isothermal wall has zero spreading resistance on its side, leaving four $k a$. The corrected prefactor is used throughout this campaign. The formulas are idealizations, especially when the contact radius is not small compared with the particle radius.

Every method uses the same conductive-material budget

Diameter [mm]	1.5	2.0	2.5
Total particles	100	300	100
High-conductivity particles	10	30	10

$$\max_{\mathbf{s}} k_{\text{eff}}(\mathbf{s}), \quad s_i \in \{0, 1\}, \quad \sum_{i \in \mathcal{B}_b} s_i = q_b$$

$s_i = 1$: conductive particle. $\mathcal{B}_b$: size class. q_b : class quota.

Fifty of 500 particles are conductive, with exactly 10% in each size class.

Every method uses the same conductive-material budget

The optimization is discrete: a particle is assigned either low or high conductivity. The constraints are exact within each size class, not just over the total count. Because all particles within a class share their diameter, fixing those counts also fixes the conductive-material volume. We do not alter positions, radii or contact topology when comparing allocations on a packing. This isolates the value of arranging the material intelligently.

Reference allocations and the search teacher

Simple reference methods

- Random: mean of 100 feasible allocations
- Degree: favour particles with many contacts
- Heat path: rank homogeneous-baseline heat throughput

Same-size swap search

- Exchange one high and one low label
- Both particles belong to the same size class
- Accept conductivity improvements
- Five restarts, 3000 proposals each

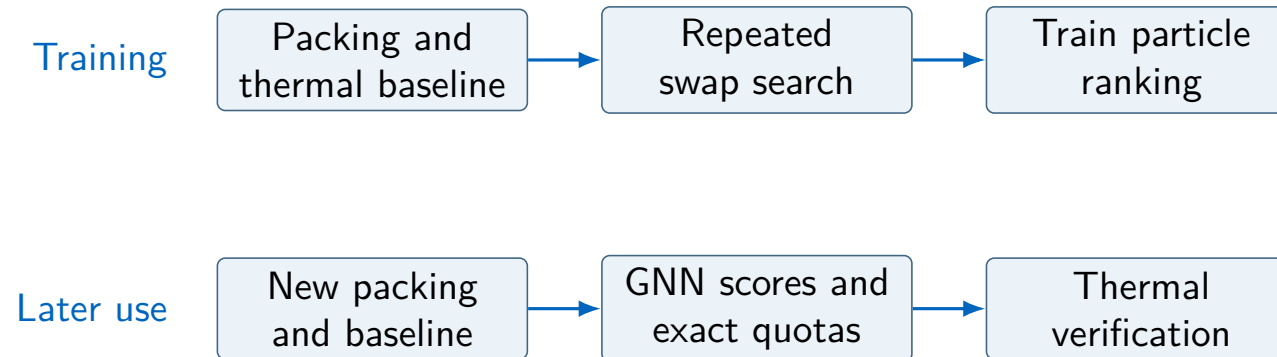
The teacher is the best allocation found within a fixed search budget.

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Reference allocations and the search teacher

Random allocation provides a distribution rather than a single lucky or unlucky baseline. Degree asks whether local connectivity alone is enough. Heat-path ranking first solves a homogeneous thermal network, then prioritizes particles carrying more heat. The search evaluates same-size swaps to preserve the quotas. Its result is a strong reference obtained with a fixed computational budget. We have not proved global optimality, so I will call it best-found search rather than the optimum.

Training and later use have different workflows



The learned ranking replaces repeated candidate search at prediction time.

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Training and later use have different workflows

Training is expensive because we solve many candidate allocations on each training packing. Those results provide supervision. Later use requires a new geometry, one homogeneous baseline solve for physical input features, a GNN evaluation, exact within-size selection and a final thermal solve. Therefore GNN scoring time alone is not total design time. In the present study, later use is emulated by withholding one packing at a time.



The GNN combines local geometry with baseline physics

Eight particle features

- Diameter, height and contact degree
- Hot-wall, cold-wall and anchored flags
- Homogeneous temperature
- Homogeneous heat throughput

Six contact features

- Contact radius
- Relative displacement in x, y, z
- Absolute vertical alignment
- Homogeneous absolute contact heat rate

Three message-passing layers, width 64. The MLP receives the same particle features, without neighbour messages.

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The GNN combines local geometry with baseline physics

The network is not asked to infer everything from geometry. A homogeneous thermal solve provides temperature and heat-throughput information that is available before assigning high-conductivity particles. The edges describe contact size, orientation and baseline heat flow. The GNN exchanges information along those edges over three layers. The MLP comparison uses the same node features, which helps isolate the contribution of neighbour information, though the GNN additionally uses the edge features. Feature scalers are fit on the training packings only.



Scores become a feasible allocation

Supervised ranking

- Search restart selections provide soft particle targets
- Listwise ranking within each size class
- Additional binary cross-entropy term
- Validation controls early stopping

Selection at prediction time

- Train three repeats per model
- Average within-class ranks
- Select the top 10 / 30 / 10 particles
- Verify conductivity with the thermal solver

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Scores become a feasible allocation

The target describes how often a particle is selected across teacher allocations. We optimize a within-class listwise ranking loss, plus binary cross entropy with weight 0.2. The model produces scores, not material conductivities or guaranteed feasible assignments. We rank within each class and select the prescribed number. Three independently trained repeats are combined through their ranks. This produces exact quotas without post-hoc changes to the material budget.



Each test packing is withheld from model fitting

28 training packings
weights and scalars

1 validation packing
early stopping

1 test packing
reported allocation

- Repeat for all 30 test packings
- Up to 400 epochs and patience 60
- Compare methods on the same held-out geometries

This tests transfer between realizations of the specified packing.

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Each test packing is withheld from model fitting

The independent unit is the packing, not an individual particle. Splitting particles from the same bed between training and testing would expose almost the same structure on both sides. Here we hold out a whole packing, use another for validation and fit on the remaining twenty-eight. We repeat the process thirty times. These folds share training data, so they are not independent training experiments. This does not establish transfer to a new porosity, particle-size distribution, system size or material contrast.

Gain and recovery answer different questions

$$\text{Gain}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{random}}}$$

$$\text{Recovery}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{search}} - k_{\text{random}}}$$

Gain

How much better is this allocation than random?

Recovery

How much of the search improvement does the model obtain?

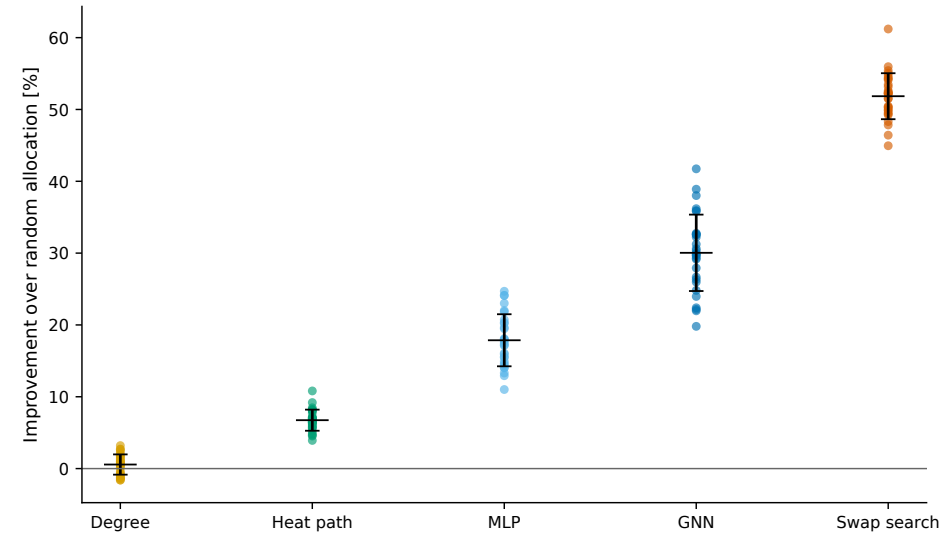
Compute both per packing, then average. Error bars describe sample SD across packings.

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Gain and recovery answer different questions

Gain normalizes the change by the random-allocation conductivity. Recovery instead divides by the available improvement found by the search. For example, a model that improves conductivity by thirty percent when search improves it by about fifty-two percent recovers roughly fifty-eight percent of the gain. The reported values are averages of per-packing ratios, so dividing displayed ensemble means will not necessarily reproduce them exactly. R-squared is not the performance measure for this discrete allocation task.

GNN improves conductivity by 30% over random



Mean improvement

Degree	0.6%
Heat path	6.7%
MLP	17.9%
GNN	30.0%
Search	51.8%

Same geometry and material budget for every comparison.

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GNN improves conductivity by 30% over random

The degree rule adds almost no average benefit. A homogeneous heat-throughput ranking is better, but both learned methods improve on it. The GNN delivers thirty percent mean gain relative to random assignment. Search still achieves substantially more, at fifty-one point eight percent. Each point is a packing, and the black summary marks show the mean and sample standard deviation. These results support learning useful allocation structure while also showing room for improvement.

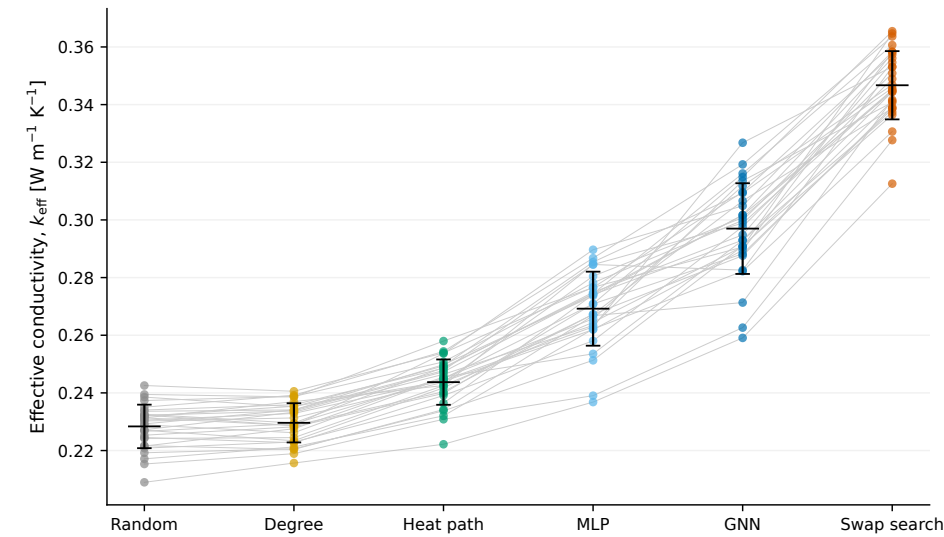
Absolute conductivity remains below the search reference



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Absolute conductivity remains below the search reference

This slide gives the absolute scale of the conductivity changes. The mean rises from approximately 0.228 watts per metre kelvin for random assignment to 0.297 for the GNN. Search reaches approximately 0.347. The thin connecting lines preserve pairing across methods. These absolute numbers belong to the corrected contact-only model and should not be mixed with values from the earlier manuscript, which used a different thermal prefactor and packing-generation implementation.

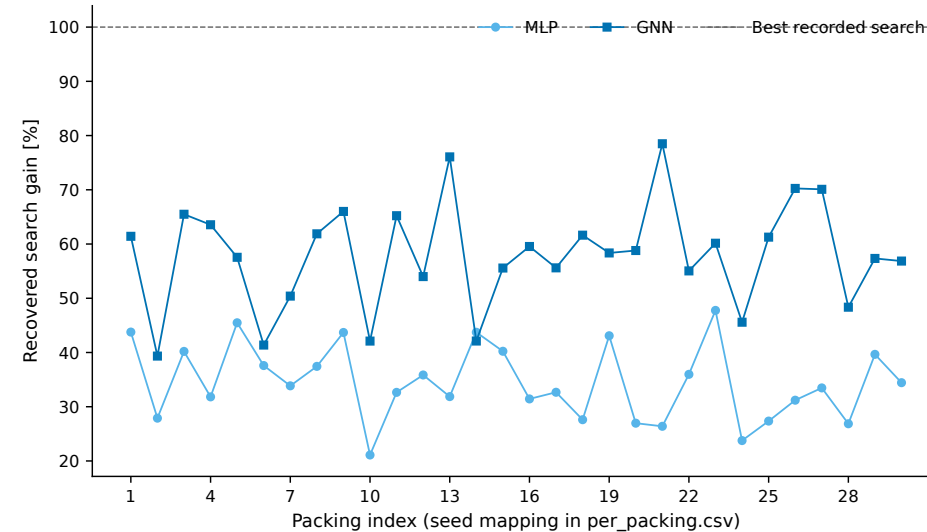


Mean k_{eff} [W/(m K)]

Random	0.2284
MLP	0.2692
GNN	0.2970
Search	0.3467

Grey lines connect results on the same packing.

GNN beats MLP on 29 of 30 held-out packings



Recovered search improvement

GNN: **58.0%**

SD: 9.8 percentage points

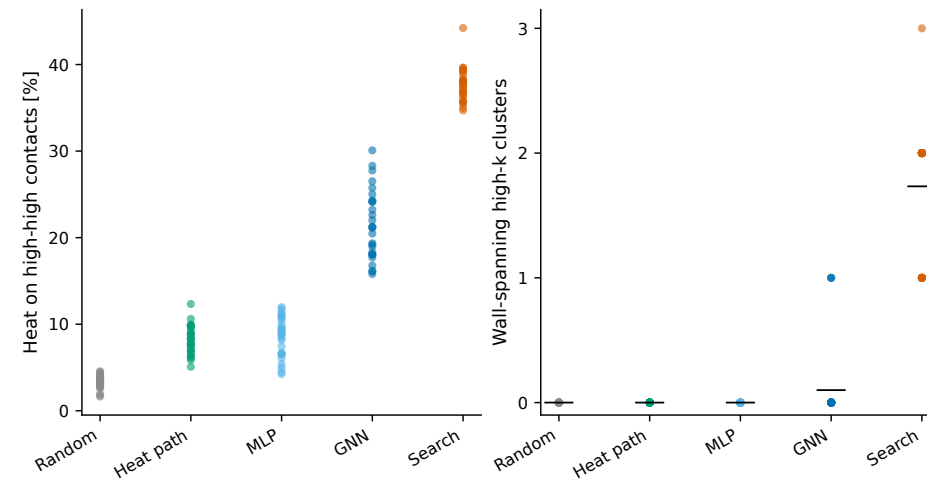
MLP: 34.5% mean

Improvement is consistent, but performance varies between packings.

GNN beats MLP on 29 of 30 held-out packings

The GNN outperforms the MLP on twenty-nine of the thirty withheld packings. Its mean recovery is fifty-eight percent, with a standard deviation of nine point eight percentage points. This is descriptive evidence across our realizations, not a claim of independent folds or universal superiority. The gap to the search reference is still substantial. Both successful and less successful packings are included in this plot.

Contact-network diagnostics help interpret allocation



Two diagnostics

- Fraction of absolute contact heat carried by high–high contacts
- Number of high-conductivity clusters spanning both walls

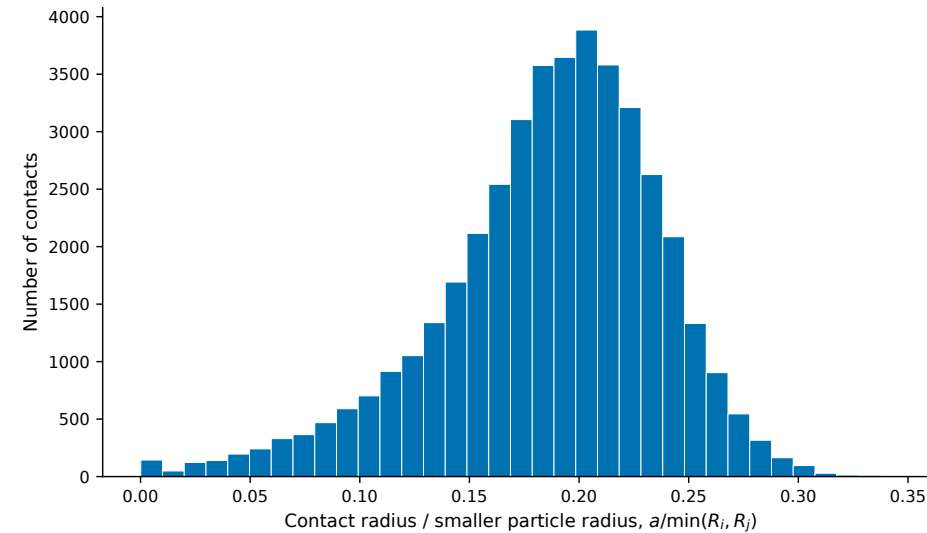
These describe the selected network. They do not prove causality.

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Contact-network diagnostics help interpret allocation

The first diagnostic measures what fraction of summed absolute particle-contact heat rates occurs on contacts whose endpoints are both high-conductivity particles. This is not the fraction of net wall heat because heat can traverse multiple contacts. The second diagnostic asks whether the high-conductivity subgraph alone spans the two thermal boundaries. Low-conductivity particles can still participate in useful paths, so a spanning high-only cluster is not required for heat transfer. The random visualization uses the sampled assignment closest to the random mean.

Numerical checks pass, but contact sizes need scrutiny



Recorded checks

- Maximum final KE:
 $8.4 \times 10^{-13} \text{ J}$
- Maximum energy imbalance:
 9.01×10^{-12}

Contact radius / smaller radius

Median: 0.192

95th percentile: 0.258

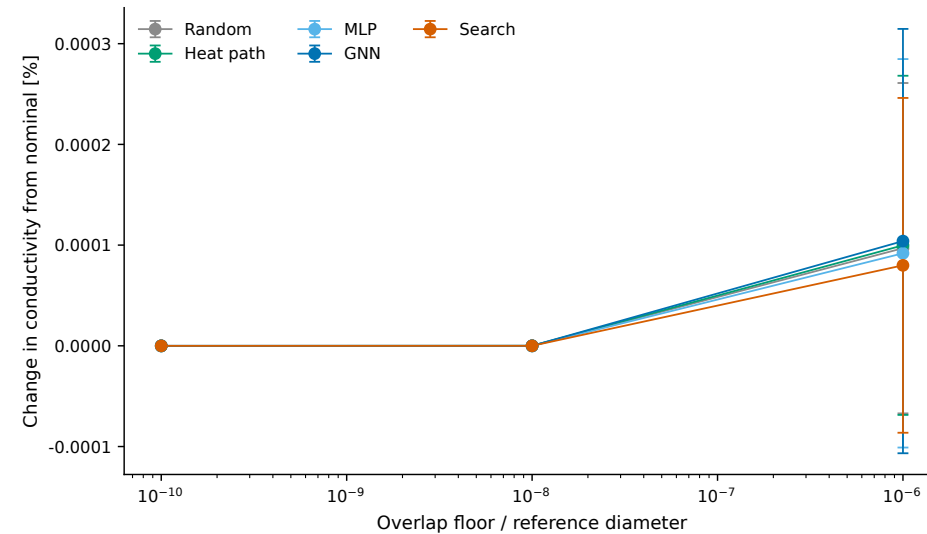
Maximum: 0.347

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Numerical checks pass, but contact sizes need scrutiny

The conservation and mechanical acceptance checks pass. However, numerical convergence does not establish physical validity. The contact-radius ratio has a median of about 0.19 and reaches about 0.35. Those contacts are not uniformly small relative to the particles. The half-space constriction approximation and the softened mechanical generation therefore deserve further assessment. This pooled histogram is a geometry diagnostic; individual contacts from the same bed are correlated.

The numerical overlap floor has little influence



Fixed allocations

- Vary the numerical overlap floor
- Largest absolute conductivity change: 0.000923%
- Sensitivity is negligible over the tested range

This does not test stiffness, loading or alternative contact laws.

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The numerical overlap floor has little influence

A small overlap floor regularizes contact-radius evaluation. We repeat the thermal evaluation with different floor values while keeping the material assignments fixed. The largest absolute relative change is less than one thousandth of a percent. That is reassuring about this numerical choice. It does not resolve the physical contact-size concern on the previous slide, because we did not regenerate the beds under another modulus, pressure or contact model.



Fast scoring still requires two thermal solves

Recorded computational work

- Search: about 14,964 thermal solves per packing
- GNN ensemble scoring: 8.63 ms mean
- MLP ensemble scoring: 0.295 ms mean

Complete prediction procedure

- Homogeneous thermal baseline
- Feature construction and model scoring
- Quota selection
- Thermal verification

The campaign demonstrates fewer candidate evaluations, not a measured end-to-end speedup.

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Fast scoring still requires two thermal solves

The repeated search requires nearly fifteen thousand thermal solves per packing. GNN scoring itself is fast, averaging about eight point six milliseconds for the ensemble. But that timing excludes the baseline and final verification solves, and it does not represent total execution time. Training and teacher-data generation are additional upfront costs. A fair future timing study should include feature construction and both solves, and should report how many later allocations are needed to amortize training.

Conclusions and next research steps

What the campaign establishes

- Material placement strongly affects modelled conductivity
- GNN gains 30% over random allocation
- Neighbour information improves on the MLP
- 58% of best-found search gain is recovered

What remains to establish

- Physical sensitivity of contact areas
- Transfer to other packing specifications
- GNN-initialized search versus random starts
- Complete runtime and manufacturing feasibility

A learned particle ranking captures useful thermal-network structure.

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Conclusions and next research steps

The main contribution is a constrained allocation workflow that combines a physical contact network with graph learning. Thirty independent packings give a clearer assessment than the earlier ten-case study. The results are promising, but the physical modelling assumptions and the remaining search-performance gap should shape the next work. A practical next algorithmic test is to initialize a short swap search with the GNN selection and compare its gain per thermal solve against random initialization. Physical sensitivity and external transfer should be evaluated before making broader claims.